

Research Reference: StarForth Physics Runtime
Volume III of the StarshipOS Technical Reference

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Chapter 1

Published Paper: Steady-State Convergence

1.1 Citation

The primary experimental results are published in:

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The paper is available at `docs/working/papers/published/James_Steady-State_Convergence_Adapt`

1.2 Abstract

1.3 Key Findings

1.4 Peer Review Notes

Chapter 2

Mathematical Companion

2.1 Q48.16 Fixed-Point Arithmetic

2.2 Feedback Loop Equations

2.3 Figures

Figure 2.1: Phase-space trajectory (snake plot).

Chapter 3

Formal Claims, Anti-Claims, and Null Hypothesis

3.1 Formal Claim Table

3.2 Anti-Claims

3.3 Null Hypothesis and Falsification Criteria

3.4 Academic Wording Guidelines

Chapter 4

Reproducibility and Independent Replication

4.1 One-Command Reproduction

4.2 Replication Invitation

4.3 Scientific Defense Checklist

4.4 Sensitivity Analysis Design

Chapter 5

Roadmap

5.1 Timeline Summary

Table 5.1: StarshipOS development roadmap

2025 (done)	2026	2027	2028
StarForth VM	StarKernel	StarshipOS	FPGA Hardware
FORTH-79 VM done	Bare metal	Self-hosting	Custom silicon
	M0–M6 done	Storage, net	Feasibility study
	M7 in progress	Multitask, self-compile	

5.2 LithosAnanke Roadmap

- v1.0.x — serial-only production (current)
- v1.5.0 — framebuffer VT100 terminal milestone
- v2.0.0 — StarForth SDK release

5.3 StarshipOS

5.4 FPGA / Custom Silicon